

CILFYNYDD, United Kingdom

WMO: 036140

Lat: 51.633N

Lon: 3.300W

Elev: 194

StdP: 99.02

Time Zone: 0.00 (EUW)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.1	-2.6	-6.2	2.3	-2.5	-4.7	2.6	-1.1	14.8	8.1	13.2	7.3	2.1	70	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	25.5	18.0	23.4	16.8	21.5	16.1	18.8	23.8	17.6	21.7	16.7	20.1	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.5	20.1	16.2	11.8	18.8	15.4	11.2	17.8	54.0	24.0	50.3	22.0	47.5	20.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.9	8.7	DB	-6.3	27.9	2.3	2.3	-7.9	29.6	-9.3	31.0	-10.5	32.3	-12.2	34.0	(4)
(5)			WB	-6.9	20.6	2.3	1.1	-8.5	21.4	-9.9	22.0	-11.2	22.6	-12.8	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	4.4	3.8	6.1	7.5	11.5	13.8	15.9	15.2	12.6	9.9	6.8	5.1	(6)	
	DBStd	5.05	3.16	3.40	2.43	2.76	3.06	3.09	2.52	2.66	2.18	2.75	3.05	3.20	(7)	
	HDD10.0	871	173	173	123	86	17	3	0	0	3	37	102	154	(8)	
	HDD18.3	3277	432	406	381	325	212	142	83	104	173	262	347	411	(9)	
	CDD10.0	658	0	0	1	11	63	116	184	163	81	33	5	2	(10)	
	CDD18.3	22	0	0	0	0	1	4	9	8	1	0	0	0	(11)	
	CDH23.3	183	0	0	0	0	7	38	72	67	0	0	0	0	(12)	
	CDH26.7	26	0	0	0	0	0	4	7	14	0	0	0	0	(13)	
(14) Wind	WSAvg	3.2	4.0	3.9	4.1	3.4	2.9	2.6	2.6	2.4	2.3	3.0	2.9	4.0	(14)	
(15) Precipitation	PrecAvg	1304	155	108	85	83	86	77	80	96	97	140	144	154	(15)	
	PrecMax	1585	266	264	174	202	169	194	218	231	199	260	280	291	(16)	
	PrecMin	881	11	10	21	12	10	14	18	16	19	41	68	41	(17)	
	PrecStd	184	67	70	44	48	39	43	47	52	45	61	55	75	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.4	14.7	20.3	24.4	26.9	27.7	28.6	22.2	17.6	13.4	12.2	(19)	
		MCWB	9.9	9.8	10.6	14.2	16.3	18.2	18.5	19.8	16.3	14.0	11.4	11.1	(20)	
	2%	DB	10.2	10.3	11.8	16.9	22.0	24.2	25.4	25.2	19.3	15.6	12.7	11.6	(21)	
		MCWB	9.3	9.0	8.9	11.6	15.3	16.8	17.9	18.5	15.1	13.3	11.7	10.7	(22)	
	5%	DB	9.5	9.3	10.2	13.8	19.5	22.1	23.2	22.0	17.8	14.7	12.0	10.8	(23)	
		MCWB	8.6	8.3	8.5	10.2	13.9	16.0	16.8	16.7	14.6	12.7	11.0	10.0	(24)	
	10%	DB	8.8	8.5	9.4	12.0	17.3	19.7	21.1	20.0	16.6	13.9	11.2	10.0	(25)	
		MCWB	7.9	7.5	8.1	9.3	12.7	14.8	16.2	15.8	13.8	12.3	10.2	9.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	10.4	11.3	14.6	16.9	19.3	19.7	20.9	17.2	14.9	12.9	11.6	(27)	
		MCDB	10.6	10.9	13.5	19.4	22.7	24.3	24.6	25.7	19.4	16.4	12.9	11.8	(28)	
	2%	WB	9.6	9.2	9.7	12.2	15.6	17.6	18.7	19.0	16.2	14.0	12.0	10.9	(29)	
		MCDB	10.0	9.9	10.9	15.3	21.2	22.6	23.7	24.2	18.3	15.0	12.4	11.3	(30)	
	5%	WB	8.8	8.6	9.1	11.0	14.4	16.4	17.6	17.6	15.2	13.2	11.2	10.1	(31)	
		MCDB	9.4	9.2	9.9	12.9	19.0	21.1	21.7	20.9	17.1	14.3	11.8	10.6	(32)	
	10%	WB	8.0	7.6	8.5	9.8	13.1	15.2	16.7	16.4	14.3	12.6	10.4	9.2	(33)	
		MCDB	8.7	8.4	9.3	11.6	16.4	19.2	20.0	18.9	16.2	13.7	11.1	9.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.7	5.4	6.8	7.7	7.6	7.3	7.5	6.9	5.6	4.9	4.1	(35)	
		MCDBR	4.9	4.9	6.5	10.0	11.4	11.5	11.8	11.4	9.0	6.2	4.7	3.7	(36)	
	5% WB	MCWBR	5.1	4.4	4.6	5.9	5.9	6.0	5.9	6.1	5.8	4.7	4.2	3.9	(37)	
		MCWBR	4.7	4.6	5.4	8.2	11.0	10.2	9.7	10.0	7.4	5.3	4.3	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.325	0.339	0.370	0.390	0.395	0.386	0.385	0.370	0.362	0.349	0.323	0.314	(40)	
		taud	2.344	2.376	2.294	2.260	2.284	2.343	2.350	2.407	2.425	2.429	2.400	2.320	(41)	
	Ebn at Noon		675	779	818	845	858	869	863	858	823	757	675	617	(42)	
		Edh at Noon	64	81	105	122	125	119	116	104	92	76	60	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.77	1.46	2.59	3.99	4.91	5.32	4.97	4.11	3.10	1.75	0.93	0.60	(44)	
	RadStd		0.07	0.18	0.33	0.46	0.25	0.57	0.48	0.36	0.23	0.16	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page